Sahaaya – Smart Healthcare Navigation System

Your Health. Our Support.

Digital Twin Challenge – prototype and code submission

Proof-of-concept prototype · all hospital data is simulated demo data

Problem Statement

Hard to choose

- Which nearby hospital offers the department or service you need?
- Information is scattered across calls, walk-ins and separate websites

Crowding is invisible

- Patients cannot see how busy a hospital is before travelling
- Long waits and wasted trips

Slow, unclear access

- Booking, tokens and directions live in different places
- Language barriers; unclear what to do in an emergency

Sahaaya brings navigation, queue visibility, booking and emergency guidance into one simple, multilingual experience.

General problem framing based on common patient experience; no survey data was collected for this prototype.

Healthcare Use Case and Target Users

Use case: patient navigation and visit planning

- Find a hospital or department by name, service or a simple sentence
- See a simulated picture of crowding (Digital Twin) before travelling
- Book an appointment and receive a demo token; track or cancel it
- Get Google Maps directions and one-tap access to 112
- Ask the local AI assistant simple questions in the chosen language

Target users

- Patients and families
- People who prefer Telugu, Hindi, Tamil or Kannada
- Caregivers planning a visit
- Future (not built): hospital front-desk and administrators

Current Approaches and Gaps

How people usually find care today

- Phone calls and walk-ins
- Individual hospital websites or apps
- Paper or counter tokens on arrival
- Word of mouth

Gaps Sahaaya targets

- No single view across hospitals
- No visibility of crowding before travelling
- Booking, token and directions are separate steps
- Limited language support and unclear emergency guidance

General observation, not a market survey or a comparison with specific products.

The Sahaaya Solution

1 Search

Hospitals, departments and services; works with plain sentences

2 Understand

Digital Twin shows simulated queue, wait and beds

3 Book

Appointment and token saved in the database

4 Go

Google Maps directions; one-tap 112 for emergencies

Sahaaya Assistant (local AI) answers questions from the prototype data, in English, Telugu, Hindi, Tamil or Kannada. Emergency messages always get a fixed "Call 112" warning – the AI is never asked.

Key Features

11 screens

- Home to Multi-Language, one flow

Smart search

- Backend search incl. sentences

Accounts

- Signup/login, scrypt-hashed passwords

Booking + tokens

- Saved in SQLite; cancel frees token

Digital Twin

- Simulated queue, beds, doctors

Local AI

- LM Studio, no API keys

Emergency

- tel:112 + safety reply

Voice + 5 languages

- Browser speech recognition

Digital Twin Prototype – Simulation Concept

SIMULATED (backend/twin.js)

- Walk-in queue: daily demand curve (peaks near 11 AM and 6 PM) plus smooth variation
- Beds free and doctors on duty
- Estimated wait = people waiting x minutes per patient / doctors
- Crowding level: Low, Moderate or High

REAL (from the database)

- Tokens booked through Sahaaya today are counted and added to the queue
- Appointments, tokens and accounts
- Not registered with any hospital

Replaceable by design: a real deployment would swap the simulate() function for verified hospital feeds without changing the rest of the app.

Labelled SIMULATED in the app. No machine-learning model is used for the twin.

AI Assistant – Details

Framework and models

- LM Studio local server (OpenAI-compatible, localhost:1234)
- Tried in order: gemma-3-1b-it, qwen3.5-0.8b, llama-3.2-1b-instruct
- No API keys; no cloud calls
- No training or fine-tuning

How it answers

- System prompt sets rules: use only given facts, no diagnosis, no medicines
- Hospital facts and twin numbers from the database are added as context
- Short answers, temperature 0.3
- Natural-language search as keyword fallback

Safety and fallbacks

- Emergency phrases (5 languages): fixed Call 112 reply before any model
- Every AI answer labelled AI-generated; can make mistakes
- AI off: keyword answer or automatic summary, no crash
- Small models: weak in Telugu, Tamil, Kannada

Technical Stack

Frontend

- HTML, CSS, vanilla JavaScript
- Tailwind CSS and Font Awesome (CDN)
- Web Speech API

Backend

- Node.js 22.13+ and Express 4
- REST JSON API
- scrypt password hashing

Data

- SQLite via built-in node:sqlite
- Tables: users, sessions, hospitals, departments, doctors, appointments, tokens

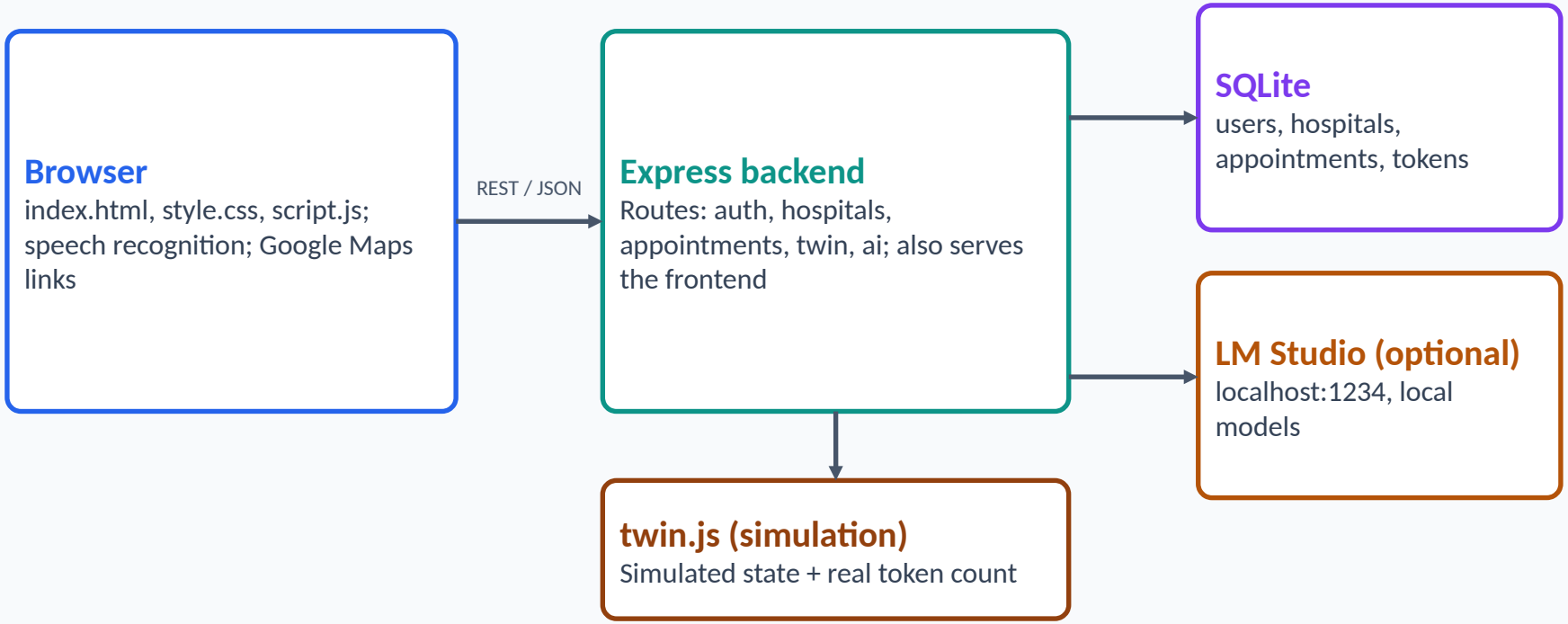
AI

- LM Studio local server
- Gemma 3 1B, Qwen 3.5 0.8B, Llama 3.2 1B (optional)

Maps and services

- Google Maps direction links (no API key)
- No paid APIs; no API keys anywhere

System Architecture (as implemented)



Full diagram: docs/Sahaaya-Architecture-Diagram.pdf. LM Studio is called only by the backend on localhost.

Demo Flow

1. Home: search "Which hospital has cardiology?"
2. Hospital Details: Digital Twin card
3. Explain with AI (labelled AI-generated)
4. Sign up / log in
5. Book appointment, receive token
6. Appointments: token status, cancel
7. Assistant; type "chest pain" for the 112 warning
8. Switch language; Google Maps directions; AI-off fallback

Expected and Observed Outcomes

Expected (hypotheses, not measured)

- Faster discovery of the right hospital or department
- Fewer wasted trips when crowding is visible
- A clearer emergency path
- Better access for local-language users

Observed in prototype testing

- All 11 screens, booking, token and cancel work; data survives a server restart
- Local AI answered in about 5 seconds on the developer laptop (informal observation)
- Emergency messages always show the 112 warning, without calling the model
- With the AI off, the app falls back gracefully
- No real-world pilot or user study yet

Limitations and Future Scope

Limitations

- Hospital data, waiting time, beds, doctors and token availability are simulated
- Not connected to any live hospital system
- Small local models make mistakes
- Interface translations cover main labels only
- Map picture is a labelled demo; no live GPS or traffic

Future scope (not implemented)

- Real hospital data from verified sources
- OpenStreetMap-based location services
- Real hospital APIs and real-time availability where officially available
- Better multilingual voice
- Production authentication and security
- Scalable cloud deployment

Demo-Data Disclaimer

Demo data – hospital availability, waiting time and token information are simulated for this prototype.

- Hospital names, doctors, addresses and timings are sample data (Bhimavaram demo set)
- Waiting time, beds, doctors on duty and token availability are simulated
- Tokens exist only inside Sahaaya and are not registered with any hospital
- AI explanations are generated from this prototype data and can be wrong
- Not connected to live hospital databases. Not medical advice. In an emergency call 112.

Team and Institution

Team details

- Team name: [FILL IN]
- Member 1 and role: [FILL IN]
- Member 2 and role: [FILL IN]
- Contact email: [FILL IN]

College / Incubator

- College or institution: [FILL IN]
- Department / programme: [FILL IN]
- City, state: [FILL IN]
- Incubator (if any): [FILL IN]